

Sandeep Singh R

PMP® Certified Technical Project Manager — Electronics / Embedded Product Development
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PROFESSIONAL SUMMARY

PMP-certified Project Manager and XLRI-certified Product Manager with 10+ years of cross-functional experience spanning Embedded Systems, IoT, Semiconductor, and Automation domains. Proven track record in full project lifecycle delivery, Agile/Scrum execution, stakeholder management, and product roadmap ownership. Adept at leading engineering teams, managing budgets and risk, and driving on-time delivery with measurable quality outcomes. Equipped with a solid foundation in technical product thinking — including PRD/user-story definition and GTM alignment — making a natural fit for Junior Product Manager opportunities alongside senior Project Manager roles. Experienced in regulatory compliance (ISO, DO-178, MISRA C) and certified in both project and product disciplines.

CORE COMPETENCIES

Project Management	Agile & Scrum, PMP, PDLC, Program Management, Risk Register, Change Control, Milestone Tracking, Scope & Schedule Management, Stakeholder Management, Budgeting & P&L, Resource Allocation, Vendor Negotiation
Product Management	Product Roadmap, User Stories, PRD/MRD, Feature Prioritization, OKRs/KPIs, Go-to-Market (GTM), Customer Discovery, Voice of Customer (VoC), Competitive Analysis, Product-Market Fit
Standards & Compliance	DO-178C (Software), MISRA C, ISO 9001:2015, BS 7262, IEC 60068, IEC, IS (Indian Standards), UL; Requirements-to-Design Traceability, V&V Methodologies, FMEA, Design Reviews
Process & Quality	Workflow Automation, QA & Validation, ISO / DO-178 / MISRA C Compliance, Design Reviews, Technical Documentation, CI/CD, Version Control
Tools & Platforms	MATLAB/SIMULINK, STM32, PIC MCU, FreeRTOS, JIRA, Git/Bitbucket, CI/CD, Agile Boards, IAR Workbench, MPLABX
Leadership	Team Building, Cross-Functional Collaboration, Mentoring, Workshop Facilitation, Client Engagement, Stakeholder Management, Customer Communication, Technical Documentation, Design Review Facilitation
Electrical/Electronic Systems	Control Systems, Firmware Integration, PCB Design & Validation, Hardware-Software V&V, Embedded Architecture, Manufacturing Readiness, FAT/Inspection Coordination

PROFESSIONAL EXPERIENCE

Project Manager | Klug Avalon Mechatronics Pvt. Ltd.

Jun 2023 – Present

Domain: Safety-Critical Embedded Systems, IoT, Electro-Mechanical Products

- Owned and Led end-to-end project lifecycle for Safe Load Indicator (SLI) embedded systems — requirements definition, hardware/firmware design, system integration, qualification testing, and formal client sign-off — directly mirroring LRU development programme workflows for safety-critical applications.
- Owned PDR and CDR equivalent design review cadences, risk register management, and milestone scheduling across 5+ concurrent projects, achieving 100% on-time delivery.
- Reduced crane safety incidents by 40% through rigorous V&V processes, structured calibration protocols, and formal design reviews aligned to safety-critical standards.
- Managed P&L oversight including vendor negotiation, component sourcing, and procurement cost optimization — maintaining BOM compliance and budget targets.
- Drove cross-functional coordination across hardware design, firmware development, and QA validation teams — improving engineering workflow efficiency by 30% and reducing inter-team handoff delays.
- Achieved 95% client satisfaction through structured milestone reviews, clear deliverable sign-offs, and proactive stakeholder communication throughout the project lifecycle.

- Maintained change control discipline aligned to PMO standards; tracked and reported KPIs including cost, schedule, quality, and delivery milestones.

Assistant Manager – R&D | *MPA Pvt. Ltd. May 2022 – Jun 2023*

Domain: Embedded Product Development, Firmware Engineering

- Cut development cycles by 20% by implementing Agile practices, sprint retrospectives, and embedded system optimization techniques.
- Boosted device performance by 25% through targeted firmware enhancements and efficient hardware resource utilization.
- Maintained 95% client satisfaction by acting as the bridge between product and engineering teams — defining requirements, managing scope, and aligning delivery timelines.
- Improved vendor coordination and component sourcing efficiency by 15% through structured RFQ processes and supplier SLAs.

Senior Embedded Design Engineer | *STPL*

Domain: Embedded Hardware & Software Integration, System Validation *Mar 2021 – May 2022*

- Drove hardware and software module design and integration for embedded products — managing PCB design validation, firmware integration, and structured design reviews consistent with CDR and integration test phases of LRU development.
- Improved system stability by 30% through systematic hardware/software integration supported by formal design reviews and structured test frameworks.
- Reduced product defects by 40% by implementing enhanced QA processes and validation frameworks consistent with DO-178/DO-254 V&V methodologies.
- Translated customer requirements into hardware/firmware feature specifications and delivery milestones — fulfilling the requirements-to-design traceability role essential in LRU programmes.

Career Break | *Nov 2019 – Mar 2021*

Personal bereavement (loss of father) coinciding with the COVID-19 pandemic. During this period, actively continued professional development as a freelance MATLAB and Embedded Systems Trainer/Developer — delivering online training sessions, developing client-specific MATLAB scripts and embedded prototypes.

Earlier Career

2011 – 2019

Embedded Design Engineer roles at Apsis Solutions, PESIT (Research), Nexsemi Systems (Microchip client), and Monee & Company

- Embedded hardware/firmware design, validation automation, and signal-processing research across semiconductor, defence-research, and industrial domains; 5 national + 2 international publications.

KEY PROJECTS

Safe Load Indicator (SLI) System for Cranes

End-to-end programme ownership: stakeholder requirements, HW/FW specification, system integration, qualification testing, and formal customer sign-off. Managed design reviews, FAT coordination, and manufacturing readiness. FMEA risk assessment identified 40% reduction in potential overload incidents through improved calibration protocols and structured V&V processes. Delivered 100% on-time across 5+ concurrent projects.

Automatic Flight Control System (AFCS)

Contributed to hardware rig design and FreeRTOS-based firmware integration for aerospace control system (defence-adjacent domain). Managed cross-functional hardware/firmware coordination, qualification testing, and formal design review artefacts. Improved system responsiveness and reduced system failures through structured hardware-software validation. Directly applicable rigorous compliance and traceability discipline to industrial control and defence projects.

Spectrometer Design & Development

Hardware: STM32 Family Microcontroller, ADCs, CCD, Flash. Software: STMCUBE, IAR Workbench. Developed STM32 firmware improving data acquisition accuracy; optimised testing methodologies reducing hardware integration time. Led requirement gathering, technical documentation, and team coordination across design, development, integration, and testing phases.

MPLAB Harmony Peripheral Library Validation

Hardware: ATMEL and PIC MCUs. Software: IAR Workbench, MPLABX, Jython. Designed test plans and test cases for peripheral library modules. Performed software simulation and hardware harness testing; identified test scenarios and edge cases. Developed and tested validation code using automated validation frameworks; generated reports and tracked issues in JIRA.

Power Smart Development Suite (GUI Dashboard)

Hardware & Software: Omicron Bode Analyzer, MATLAB/SIMULINK. Understood design requirements for GUI development and integrated calculations, populating values per specification. Tested GUI under various operating conditions; generated reports and coordinated with team on development changes. Accelerated engineering decision-making and improved visualization efficiency.

Home Automation over IoT (Falcon Controller, Kite Switch, IR Blaster, Smart Plug)

Hardware & Software: PIC24 & 32 MCU, MPLAB, Ambient sensor, LED drivers, Triac, MiWi, Ethernet, EEPROM, Flash memory. Developed embedded drivers for various MCU peripherals; performed hardware design, board bring-up, and verification with structured test cases. Integrated multiple subsystems and demonstrated IoT connectivity and control architecture.

Channel Sounding Experiments – Underwater Acoustic Channels

Hardware & Software: NI DAQ tools, MATLAB. Designed hardware electronics for data acquisition; interfaced NI-DAQ devices and microcontrollers. Conducted acoustic experiments and performed reliability testing; collected and analysed large volumes of data with high accuracy. Demonstrated outstanding project management running research programmes within cost and timeframe parameters. Published findings in journals; built partnerships with industry and academic collaborators.

Rotating Data Acquisition System – Strain Gauge Testing

Hardware: PIC32MX460/795F512L Microcontroller, ADC, MCP6N11, SST25v080B flash, RF Module, Strain gauge. Software: MPLABX, MATLAB/SIMULINK. Performed system reliability testing and performance verification; developed application and GUI using MATLAB/SIMULINK. Designed CAD models and specifications; ensured compliance with health and safety practices and engineering standards.

EDUCATION & CERTIFICATIONS

M.Tech – Signal Processing | SJCIT, VTU | 2016

B.E – Electronics & Communication | SSEC, VTU | 2011

Certifications

- PMP® – Project Management Professional (PMI)
- Product Management Certification – XLRI
- Embedded Systems & Peripherals – Microchip India Pvt. Ltd.
- Online Certifications – Project Management & Embedded Systems (Coursera)

PATENT

Automobile Emergency Alerting Autonomous Device – Design Patent Filed, India (Reg. No. 453281-001) | Mar 2025

ACHIEVEMENTS & ADDITIONAL

- Winner – Best Design, Embedded Design Contest (EFY)
- Published 5 National Conference Papers & 2 International Journals
- MATLAB Instructor – Udemy (Basic Course); conducted multiple engineering student workshops
- NCC 'B' Certificate Holder
- Passionate about adventure sports: rock climbing, rappelling, trekking